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Broca's area, variation and taxic diversity in early *Homo* from Koobi Fora (Kenya)

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Abstract

Because brain tissues rarely fossilize, pinpointing when and how modern human cerebral traits emerged in the hominin lineage is particularly challenging. The fragmentary nature of the fossil material, coupled with the difficulty of characterizing such a complex organ, have been the source of long-standing debates. Prominent among them is the uncertainties around the derived or primitive state of the brain organization in the earliest representatives of the genus *Homo*, more particularly in key areas such as the Broca's area. By revisiting a particularly well-preserved fossil endocast from the Turkana basin (Kenya) attributed to early *Homo*, here we confirm that humans in Africa had a primitive organization of the Broca's area ca. 1.9 million years ago. Additionally, our description of KNM-ER 3732 adds further information about the variation pattern of the inferior frontal gyrus in fossil hominins, with implications for early *Homo* taxic diversity (i.e., one or two *Homo* species at Koobi Fora) and the nature of the mechanisms involved in the emergence of derived cerebral traits.

eLife assessment

This **important** study uses the brain endocast of a ~1.9-million-year-old hominin fossil from Kenya, attributed to genus *Homo*, to show that the organization of the Broca's area in members of early *Homo* was primitive. Specifically, the prefrontal sulcal pattern in this early *Homo* specimen more closely resembles that of chimpanzees than of modern humans. Because Broca's area is associated with speech function, the **compelling** evidence from this study has relevant for understanding the timing and trajectory of evolution of speech related traits in our genus. Coupled with its potential implications for taxonomic classification, this study will be of interest to paleoanthropologists, paleontologists, archaeologists, and neuroscientists.

Introduction

The modern human brain is an exceptionally complex, highly specialized and extremely costly machinery. Because of the fragmentary nature of the hominin fossil record, assessing when and how changes in the brain of our ancestors happened, and inferring any related functional, behavioral and metabolic consequences, is particularly challenging (Zollikofer and Ponce de León, 2013). Nonetheless, reconstructing the chronological and taxonomical context of the emergence of derived cerebral traits is a prerequisite for disentangling underlying evolutionary processes. For instance, local reorganization of specific areas in an overall primitive hominin brain would support a mosaic-like evolutionary pattern (e.g., Holloway et al., 2004), and raise essential questions on the role of selection pressure (or absence of; rev. in Beaudet, 2021). Beyond the value of such information on the origins of the human brain, the assumption that *Homo* developed a uniquely complex brain organization (e.g., Tobias, 1987) requires further evidence.

Although brains rarely fossilize, it is possible to glean structural information about the evolutionary history of the hominin brain by studying sulcal imprints in fossil brain endocasts (Neubauer et al., 2014). In this regard, the Broca's cap has been the focus of much interest in paleoneurology due to striking structural differences between extant human and chimpanzee brains and endocasts, and the implication of this area in articulated language (rev. in Beaudet, 2017). Ponce de León et al. (2021) thoroughly examined brain endocasts of *Homo* specimens in eastern Africa and Eurasia and demonstrated that the organization of the Broca's area in the earliest representatives of the genus before 1.5 Ma was primitive. Because the imprints of the Broca's cap in endocasts are not always readable, they used the coronal suture and surrounding sulcal imprints as a proxy to identify frontal lobe expansion (i.e., derived condition). Unfortunately, in some of the oldest and key specimens from Africa that could represent the >1.5 Ma condition (e.g., KNM-ER 1470) the interpretations remain inconclusive.

Testing the hypothesis of Ponce de León et al. (2021) of a primitive brain in the earliest representatives of the genus *Homo* before 1.5 Ma thus necessitates (i) an excellent preservation of very fine neuroanatomical details in fossil endocasts, and (ii) reliable information on their taxonomic identity (i.e., *Homo*) and stratigraphic context (i.e., before/after 1.5 Ma). The hominin specimen KNM-ER 3732 from Kenya, which fulfills these requirements, has the potential to shed new light on this conundrum. KNM-ER 3732 was discovered during the 1974-1975 field programme in the Koobi Fora Formation (area 115), east of Lake Turkana in Kenya (Leakey, 1976; Wood, 1991). KNM-ER 3732 consists of a calotte, left zygoma and a natural endocast. The expanded neurocranium and robust upper face supports an attribution to *Homo* sp. indet. (Wood, 1991). The specimen was lying below the KBS Tuff of the upper Burgi Member that is dated to 1.87 million years ago, which provides a minimum age (Feibel et al., 1983, 2009). Accordingly, if KNM-ER 3732 shows a derived cerebral condition, the hypothesis of >1.5 Ma *Homo* being associated to a primitive organization of the Broca's area is falsified. Here we provide a comparative study of the natural endocast of pre-1.5 Ma *Homo* specimen KNM-ER 3732 to test the hypothesis of a late emergence of a modern Broca's cap in the hominin lineage.

Results

KNM-ER 3732 offers a glimpse of early *Homo* prefrontal organization

The natural endocast of KNM-ER 3732, initially described by Holloway et al. (2004), preserves the dorsal part but misses the frontal pole, the occipital and temporal lobes, as well as the entire ventral surface (Fig. 1). Wood (1991: 132) identified an “irregular bony excrescence on the surface of the right parietal at the level of lambda and 32 mm from the midline. It projects 6 mm from the surface of the bone, and may be an example of myositis ossificans associated with damage to the overlying temporal muscle”. Because of the location (right parietal) and nature (traumatic) of the outgrowth, it does not affect our area of study (i.e., left frontal bone) (Walczak et al., 2015). The gyral and sulcal details are well-preserved, particularly in the prefrontal region. The superior, middle and inferior frontal sulci are visible on both hemispheres. On the left hemisphere, the Broca’s cap is prominent. A vertical groove, identified as the precentral sulcus, separates the Broca’s cap in half (Fig. 2A). On both hemispheres, the central sulcus intersects the inter-hemispheric scissure and the post-central sulcus seems to be connected to the lateral fissure. The superior temporal sulcus nearly intersects the lateral fissure on the right hemisphere. The intra-parietal sulcus can be found in the parieto-occipital region of both hemispheres. The examination of the internal surface of the braincase (Fig. 2B) confirms the pattern described on the natural endocast.

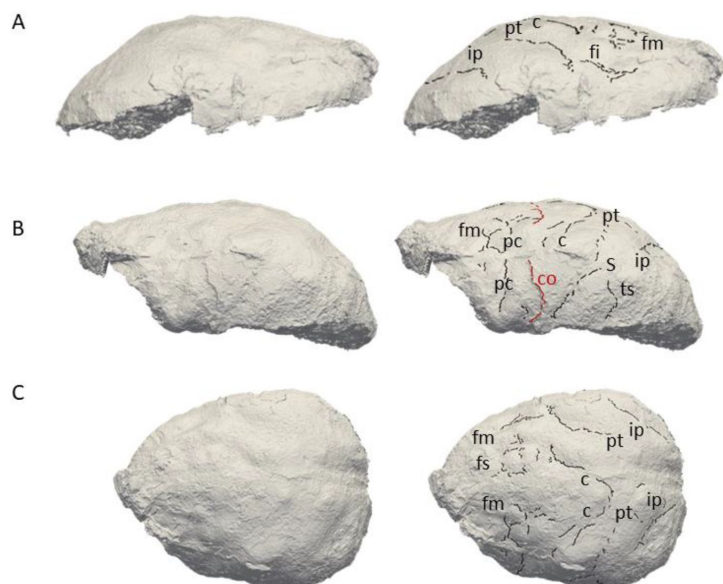


Figure 1.

3D virtual rendering of the natural endocast of KNM-ER 3732 and identification of sulcal imprints. KNM-ER 3732 is shown in lateral right (A), lateral left (B) and dorsal (C) views. Images not to scale. ar: ascending ramus of the lateral fissure; c: central sulcus; CO: coronal suture; fi: inferior frontal sulcus; fm: middle frontal sulcus; ip: intra-parietal sulcus; pc: pre-central sulcus; pt: post-central sulcus.

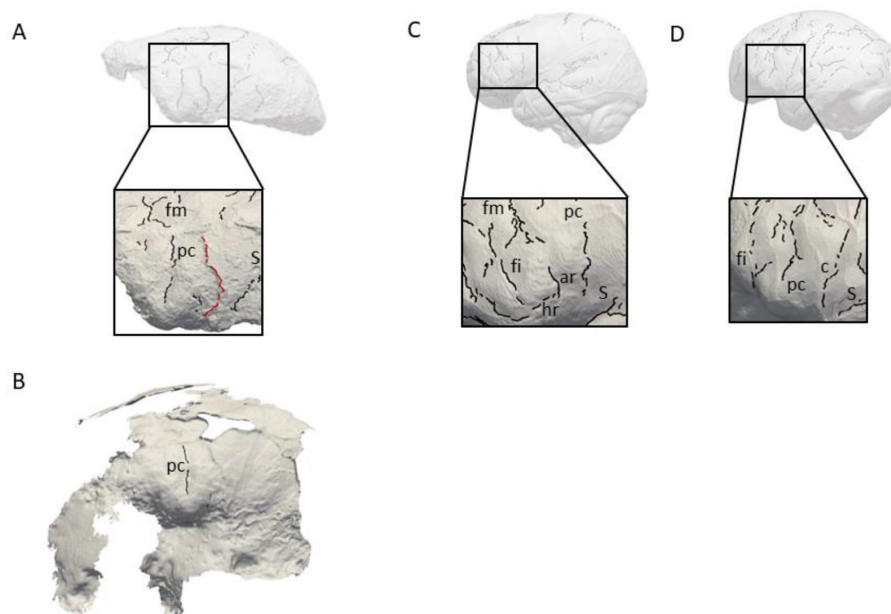


Figure 2.

Comparison of the sulcal patterns identified in the inferior frontal area of KNM-ER 3732. The natural endocast (A) and braincase (B) of KNM-ER 3732 are compared to the virtual endocasts of extant human (C) and chimpanzee (D) individuals. Images not to scale. ar: ascending ramus of the lateral fissure; c: central sulcus; co: coronal suture; fi: inferior frontal sulcus; fm: middle frontal sulcus; ip: intra-parietal sulcus; pc: pre-central sulcus; pt: post-central sulcus.

KNM-ER 3732 has a primitive prefrontal cortex

When compared to extant human and chimpanzee sulcal patterns, KNM-ER 3732 closely resembles the latter (Fig. 2C-D). In extant humans, the inferior frontal sulcus often transects the Broca's region, while the ascending ramus of the lateral fissure caudally borders the prominence (Figure 2C-D; Connolly, 1950; de Jager et al., 2022). In chimpanzees, the central sulcus is placed more rostrally, and the inferior portion of the precentral sulcus bisects the orbital cap that is bordered rostrally by the inferior frontal sulcus (Figure 2E-F; Connolly, 1950). The coronal suture runs in between the precentral and central sulci in KNM-ER 3732, which points towards the primitive configuration described in Ponce de León et al. (2021). The sulcal pattern seen in the prefrontal cortex of KNM-ER 3732, and more particularly the deeply marked precentral sulcus that incised the Broca's cap, is not found in any of the early *Homo* specimens described in Ponce de León et al. (2021) but approximates the condition seen in the Dmanisi cranium D2282 (Fig. S1B in Ponce de León et al., 2021). However, this pattern is seen in a contemporaneous non-*Homo* hominin specimen in South Africa, i.e., *Australopithecus sediba* (Falk, 2014).

Discussion

Paleoneuroanatomy supports taxic diversity within early *Homo*

Overall, the present study supports Ponce de León et al.'s (2021) hypothesis of a primitive brain of early *Homo* but also adds information about the variation pattern of the inferior frontal gyrus. In particular, the diversity of the prefrontal sulcal pattern of hominin endocasts at Koobi Fora revives the debate about the possible presence of two early *Homo* species in this locality (i.e., *Homo habilis sensu stricto* and *Homo rudolfensis*). In 1983, Falk (1983) published the description of the endocasts of two *Homo* specimens from Koobi Fora, KNM-ER 1470 and KNM-ER 1805. Her analysis supported the co-existence of two morphs, KNM-ER 1470 representing a more derived human-like sulcal pattern. Interestingly, cerebral evidence brought up by her analysis matched other studies that emphasized the more

derived craniodental anatomy of specimens attributed to *Homo rudolfensis* as opposed to the more primitive (*Australopithecus*-like) traits identified within *Homo habilis sensu stricto* (Leakey et al., 2012). While describing the external morphology of the neurocranium, Leakey (1976: 575) noted that KNM-ER 3732 was “strikingly similar to KNM-ER 1470”. This resemblance is not reflected in their cerebral organization since the present study rather suggests a primitive organization of the Broca’s cap in KNM-ER 3732. If KNM-ER 1470 had indeed a derived brain, taxic diversity as a source of variation cannot be discarded. If we go further down that route, the similarities between KNM-ER 3732 and *Australopithecus sediba* revealed by our study could be an argument supporting the presence of *Australopithecus* in Koobi Fora (Wood and Collard, 1999).

The evolutionary history of the human Broca’s area unravelled

Beyond the taxonomic aspect, variation detected in Koobi Fora could provide information about underlying evolutionary mechanisms, and more specifically the process of fixation of an adaptive variant, i.e., a new organization of the Broca’s cap and the increase of neural interconnectivity in this region (Van Essen, 1987). Such neurological changes might have had deep implication in the emergence of novel behavior, such as articulated language (Beaudet, 2017, 2021). Within this scenario, the study from Ponce de León et al. (2021) would suggest that this trait became stabilized by 1.5 million years ago. The identification of protracted brain growth as early as in *Australopithecus afarensis* (Gunz et al., 2020), inducing longer exposure to social environment during brain maturation (rev. in Hublin et al., 2014), could be consistent with a derived organization of the Broca’s cap being selected as a response to social environmental stimuli through developmental plasticity, culminating in this variant becoming dominant within *Homo*.

Materials and Methods

Materials

KNM-ER 3732 is currently housed in the National Museums of Kenya in Nairobi (Kenya). Comparative extant human (N = 1, adult female) and chimpanzee (N = 1, adult female) specimens are from the Pretoria Bone Collection at the University of Pretoria (South Africa) and in the Royal Museum for Central Africa in Tervuren (Belgium), respectively (Beaudet et al., 2018).

Digitization

KNM-ER 3732 was scanned using an Artec Space Spider 3D scanner and reconstructed with the software Artec Studio 16X. The 3D mesh can be viewed on MorphoSource (<https://www.morphosource.org/concern/media/000497752?locale=en>). The comparative specimens were imaged by microfocus X-ray tomography (21) at the South African Nuclear Energy Corporation in Pelindaba (South Africa) and at the Centre for X-ray Tomography of Ghent University (UGCT) in Ghent (Belgium). Virtual endocasts were generated using Endex software (Subsol et al., 2010).

Detection and identification of sulcal imprints

Sulcal imprints were automatically detected through a geometry-based method using curvature lines computed from the surface. Sulcal imprints, considered as variation points of the surface on a triangle mesh, were detected through a geometry-based method using curvature lines defined as salient subsets of the extrema of the principal curvatures on surfaces (Yoshizawa et al., 2008; Beaudet et al., 2016). Vascular imprints and non-anatomical structures (e.g., fractures) were manually removed through a customized script written in MATLAB R2013a (Mathworks).

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We thank Emmanuel Ndiema and the curatorial staff of the National Museums of Kenya for collection access. For scientific/technical discussion we are grateful to: G. Castelli (Cambridge), M. Mirazón-Lahr (Cambridge), R. Holloway (New York), F. Spoor (London). AB is funded by the National Research Foundation of South Africa (Research Development Grants for Y-Rated Researchers, grant number 129336) and the University of Cambridge. EdJ is funded by the University of Cambridge Harding Distinguished Postgraduate Scholars Programme. AB and EdJ are funded by the South Africa/France (PROTEA) Joint Research Programme (grant number 129923) and the McDonald Institute for Archaeological Research.

Competing Interest Statement

The authors declare no competing interest.

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Reviewer #1 (Public Review):

The cerebral cortex, or surface of the brain, is where humans do most of their conscious thinking. In humans, the grooves (sulci) and bumps (convolutions) have a particular pattern in a region of the frontal lobe called Broca's area, which is important for language. Specialists study features imprinted on the internal surfaces of braincases in early hominins by casting their interiors, which produces so-called endocasts. A major question about hominin brain evolution concerns when, where, and in which fossils a humanlike Broca's area first emerged, the answer to which may have implications for the emergence of language. The researchers used advanced imaging technology to study the endocast of a hominin (KNM-ER 3732) that lived about 1.9 million years ago (Ma) in Kenya to test a recently published hypothesis that Broca's remained primitive (apelike) prior to around 1.5

Ma. The results are consistent with the hypothesis and raise new questions about whether endocasts can be used to identify the genus and/or species of fossils.

Reviewer #2 (Public Review):

The authors tried to support the hypothesis that early Homo still had a primitive condition of Broca's cap (the region in fossil endocasts corresponding to Broca's area in the brain), being more similar to the condition in chimpanzees than in humans. The evidence from the described individual points to this direction but there are some flaws in the argumentation.

First, only one human and one chimpanzee were used for comparison, although we know that patterns of brain convolutions (and in addition how they leave imprints in the endocranial bones) are very variable.

Second, the evidence from this fossil specimen adds to the evidence of previously describe individuals but still not yet fully prove the hypothesis.

Third, there is a vicious circle in using primitive and derived features to define a fossil species and then using (the same or different) features to argue that one feature is primitive or derived in a given species. In this case, we expect members of early Homo to be derived compared to their predecessors of the genus *Australopithecus* and that's why it seems intriguing and/or surprising to argue that early Homo has primitive features. However, we should expect that there is some kind of continuum or mosaic in a time in which a genus "evolves into" another genus. This discussion requires far more discussions about the concepts we use, maybe less discussion about what is different between the two groups but more discussion about the evolutionary processes behind them.

Fourth, the data of convolutional imprints presented are rather subjective when identifying which impressions represent which brain convolutions. Not seeing an impression does not necessarily mean that the corresponding brain feature did not exist. Interestingly, the manuscript does not mention and discuss at all the frontoorbital sulcus. This is a sulcus that usually runs from the orbital surface of the frontal lobe up to divide the inferior frontal gyrus in chimpanzees, a condition totally different than in humans who do not have a frontoorbital sulcus. Could such a sulcus be identified, this would provide a far more convincing argument for a primitive condition in this specimen. In *Australopithecus sediba*, e.g., the condition in this region seems to be a mosaic in which some aspects of the morphology seem to be more modern while one of the sulcal impressions can well be interpreted as a short frontoorbital sulcus. For this specimen, by the way, I would come back to my third point above: some experts in the field might argue that this specimen could belong to Homo rather than *Australopithecus*...

According to my arguments above, I think that this manuscript might revive interesting discussions about this topic but it is not likely to settle them because the data presented are not strong enough to fully support the hypothesis.

Reviewer #3 (Public Review):

The authors provide a detailed analysis of the sulcal and sutural imprints preserved on the natural endocast and associated cranial vault fragments of the KNM-ER3732 early Homo specimen. The analyses indicate a primitive ape-like organization of this specimen's frontal cortex. Given the geological age of around 1.9 million years, this is the earliest well-documented evidence of a primitive brain organization in African Homo.

In the discussion, the authors re-assess one of the central questions regarding the evolution of early Homo: was there species diversity, and if yes, how can we ascertain it? The specimen KNM-ER1470 has assumed a central role in this debate because it purportedly shows a more advanced organization of the frontal cortex compared to other largely coeval specimens (Falk, 1983). However, as outlined in Ponce de León et al. 2021 (Supplementary Materials), the imprints on the ER1470 endocranium are unlikely to represent sulcal structures and are more likely to reflect taphonomic fracturing and distortion. Dean Falk, the author of the 1983 study, basically shares this view (personal communication). Overall, I agree with the authors that the hypothesis to be tested is the following: did early Homo populations with primitive versus derived frontal lobe organizations coexist in Africa, and did they represent distinct species?

I greatly appreciate that the authors make available the 3D surface data of this interesting endocast.

Author Response:

Reviewer #1 (Public Review):

The cerebral cortex, or surface of the brain, is where humans do most of their conscious thinking. In humans, the grooves (sulci) and bumps (convolutions) have a particular pattern in a region of the frontal lobe called Broca's area, which is important for language. Specialists study features imprinted on the internal surfaces of braincases in early hominins by casting their interiors, which produces so-called endocasts. A major question about hominin brain evolution concerns when, where, and in which fossils a humanlike Broca's area first emerged, the answer to which may have implications for the emergence of language. The researchers used advanced imaging technology to study the endocast of a hominin (KNM-ER 3732) that lived about 1.9 million years ago (Ma) in Kenya to test a recently published hypothesis that Broca's remained primitive (apelike) prior to around 1.5 Ma. The results are consistent with the hypothesis and raise new questions about whether endocasts can be used to identify the genus and/or species of fossils.

We would like to thank Rev. 1 for their comments on our paper.

Reviewer #2 (Public Review):

The authors tried to support the hypothesis that early Homo still had a primitive condition of Broca's cap (the region in fossil endocasts corresponding to Broca's area in the brain), being more similar to the condition in chimpanzees than in humans. The evidence from the described individual points to this direction but there are some flaws in the argumentation.

We are grateful to Rev. 2 for their comments, although we partially agree with some of them.

First, we would like to rectify the statement of Rev. 2 that we “tried to support the hypothesis that early Homo still had a primitive condition of Broca's cap”, indeed, our aim was to test this hypothesis and not to try to validate it.

First, only one human and one chimpanzee were used for comparison, although we know that patterns of brain convolutions (and in addition how they leave imprints in the endocranial bones) are very variable.

We understand the point raised by Rev. 2 about the variation of brain convolutions in humans and chimpanzees. We used atlases published by Connolly (1950), Falk et al. (2018) and de Jager et al. (2019, 2022) to analyse the endocast of KNM-ER 3732 and compare it to the extant

human and chimpanzee cerebral conditions. However, in Figure 2, for the sake of clarity only two Homo and Pan specimens were used to illustrate the comparison (as it has been done in other published papers, e.g., Carlson et al., 2011; Science, Gunz et al., 2020 Sci Adv). In the revised version, we modified the manuscript to explain further our approach (line 156) “We used brain and endocast atlases published in Connolly (1950), Falk et al. (2018) and de Jager et al. (2019, 2022; see also www.endomap.org) for comparing the pattern identified in KNM-ER 3732 to those described in extant humans and chimpanzees. To the best of our knowledge, these atlases are the most extensive atlases of extant human and chimpanzee brains/endocasts available to date and are widely used in the literature to explore variability in sulcal patterns. In Figure 2, the extant human and chimpanzee conditions are illustrated by one extant human (adult female) and one extant chimpanzee (adult female) specimens from the Pretoria Bone Collection at the University of Pretoria (South Africa) and in the Royal Museum for Central Africa in Tervuren (Belgium), respectively (Beaudet et al., 2018).”.

Second, the evidence from this fossil specimen adds to the evidence of previously describe individuals but still not yet fully prove the hypothesis.

We tempered our discussion by concluding that (line 116) “Overall, the present study not only demonstrates that Ponce de León et al.’s (2021) hypothesis of a primitive brain of early Homo cannot be rejected, but also adds information [...]”.

Third, there is a vicious circle in using primitive and derived features to define a fossil species and then using (the same or different) features to argue that one feature is primitive or derived in a given species. In this case, we expect members of early Homo to be derived compared to their predecessors of the genus Australopithecus and that's why it seems intriguing and/or surprising to argue that early Homo has primitive features. However, we should expect that there is some kind of continuum or mosaic in a time in which a genus "evolves into" another genus. This discussion requires far more discussions about the concepts we use, maybe less discussion about what is different between the two groups but more discussion about the evolutionary processes behind them.

We fully agree with Rev. 2 on this aspect. We believe that identifying these differences/similarities between fossil and extant hominids constitute the first step of a better understanding of the evolutionary mechanisms. Our work suggests indeed a certain continuity between genera and raises questions on the genus concept and how to interpret the specimens currently attributed to early Homo. In the revised version of the manuscript we included a reference to this possible scenario (line 134): “[...] or to the absence of a definite threshold between the two genera based on the morphoarchitecture of their endocasts (Wood and Collard, 1999).”.

Fourth, the data of convolutional imprints presented are rather subjective when identifying which impressions represent which brain convolutions. Not seeing an impression does not necessarily mean that the corresponding brain feature did not exist. Interestingly, the manuscript does not mention and discuss at all the frontoorbital sulcus. This is a sulcus that usually runs from the orbital surface of the frontal lobe up to divide the inferior frontal gyrus in chimpanzees, a condition totally different than in humans who do not have a frontoorbital sulcus. Could such a sulcus be identified, this would provide a far more convincing argument for a primitive condition in this specimen. In Australopithecus sediba, e.g., the condition in this region seems to be a mosaic in which some aspects of the morphology seem to be more modern while one of the sulcal impressions can well be interpreted as a short frontoorbital sulcus. For this specimen, by the way, I would come back to my third point above: some experts in the field might argue that this specimen could belong to Homo rather than Australopithecus...

We agree that the presence of a fronto-orbital sulcus would be more conclusive. However, this sulcus has not been identified in KNM-ER3732 and the region in which we would expect to find

it is not preserved. As demonstrated by Ponce de León et al. (2021), because of the topographic relationships between sulci (and cranial structures), it is possible to interpret imprints on endocasts and the evolutionary polarity of some traits even in the absence of landmarks such as the fronto-orbital sulcus. In *Australopithecus sediba* the main derived feature of the endocast corresponds to the ventrolateral bulge in the left inferior frontal gyrus, and not to the sulcal pattern itself (Carlson et al., 2011 Science). However, the discussion around the taxonomic status of this taxon confirms the urgent need for reconsidering specimens from that time period and clarifying the mosaic-like or concerted evolution of the derived Homo-like traits within our lineage. Regarding the subjective nature of this approach, we invite readers to examine the specimen on MorphoSource (<https://www.morphosource.org/concern/media/000497752?locale=en>) and to request access to the National Museums of Kenya to the physical or virtual specimen to falsify our hypothesis.

According to my arguments above, I think that this manuscript might revive interesting discussions about this topic but it is not likely to settle them because the data presented are not strong enough to fully support the hypothesis.

We would be more than happy to consider new/other specimens with similar chronological and geographical contexts and investigate further this hypothesis in the future.

Reviewer #3 (Public Review):

The authors provide a detailed analysis of the sulcal and sutural imprints preserved on the natural endocast and associated cranial vault fragments of the KNM-ER3732 early Homo specimen. The analyses indicate a primitive ape-like organization of this specimen's frontal cortex. Given the geological age of around 1.9 million years, this is the earliest well-documented evidence of a primitive brain organization in African Homo.

In the discussion, the authors re-assess one of the central questions regarding the evolution of early Homo: was there species diversity, and if yes, how can we ascertain it? The specimen KNM-ER1470 has assumed a central role in this debate because it purportedly shows a more advanced organization of the frontal cortex compared to other largely coeval specimens (Falk, 1983). However, as outlined in Ponce de León et al. 2021 (Supplementary Materials), the imprints on the ER1470 endocranium are unlikely to represent sulcal structures and are more likely to reflect taphonomic fracturing and distortion. Dean Falk, the author of the 1983 study, basically shares this view (personal communication). Overall, I agree with the authors that the hypothesis to be tested is the following: did early Homo populations with primitive versus derived frontal lobe organizations coexist in Africa, and did they represent distinct species?

I greatly appreciate that the authors make available the 3D surface data of this interesting endocast.

We are grateful to Rev. 3 for their comments and for contextualizing our finding. We would also like to point out that, although the 3D surface can be viewed on MorphoSource, permission from the National Museums of Kenya has to be requested for studying the specimen and getting access to the physical specimen and/or the 3D model.